

## Office Occupancy vs. WFH Trends (2019–2030)

**Measured occupancy (pre- vs. post-COVID):** Before the pandemic, office attendance was treated as 100% baseline (Feb 2020=100% occupancy) [14†L57-L60] . In practice, pre-2020 daily attendance on peak days was typically 80–100% of assigned seating. Since COVID, in-person occupancy collapsed. For example, Kastle Systems (keycard data) reports **peak-day** attendance in major US cities at only ~62% of Feb-2020 levels (Mar 2024) [14†L57-L60] , with **average weekly occupancy ~52–55%** of the same baseline [14†L75-L79] [11†L263-L270] . XY Sense sensor data similarly show current *global* workspace occupancy  $\approx$ 40% of capacity (Q4 2024–Q1 2025) [9†L7-L14] . Cushman & Wakefield and CBRE find overall utilization stabilized at ~50–60% of pre-pandemic levels by 2024 [4†L126-L134] [17†L199-L207] .

**Weekly/diurnal patterns:** All data show a pronounced midweek peak: Tuesdays (and often Wednesdays) are busiest, while Mondays and especially Fridays are much lighter. Kastle’s “hybrid peak” (Tue–Thu) average is ~58% of pre-COVID occupancy (2023) [17†L210-L218] , with Tuesdays reaching ~60% and Fridays ~30% (only ~43% below the midweek peak) [17†L222-L230] . An Axios analysis similarly finds occupancy  $\approx$ 62% on peak days vs. ~35% on Fridays (Mar 2024) [14†L75-L79] . Surveys and sensor indices (e.g. VergeSense) confirm **Tuesday–Thursday** highest usage, with Mondays 30–40% below midweek and Fridays <50% of midweek [6†L220-L223] [20†L278-L285] . Diurnal data show a two-peaked day: arrivals cluster 9–11 AM (peak ~11 AM) with a midday drop, a second peak ~2 PM, and a rapid decline after ~4 PM [20†L295-L304] .

**Measured utilization:** Most offices remain *over-desked*. Even with ~50% attendance, available workstations are far underused: e.g. XY Sense finds ~25% of desks never used and 19% used <1 hour/day [8†L144-L152] . In short, typical modern offices see 50–60% of employees in the building on the busiest days, far lower on Mondays/Fridays, and many spaces sit empty. This holds in Canada as well; while direct Canadian occupancy sensor data are scarce, Statistics Canada reports **stable** ~20% of workers WFH by late 2023 [25†L79-L88] (implying ~80% office-attendance potential), comparable to US patterns.

## 2. Canadian WFH Trends (2019–2024)

Canadian labour surveys show a huge COVID spike in homeworking. Only ~7–8% of Canadians “usually” worked mainly from home pre-pandemic (e.g. January–Feb 2020) [25†L85-L89] [27†L109-L112] . In April 2020 this leapt to ~40% (strict lockdown) and then receded as offices re-opened. By January 2022 about **30%** of workers were still doing most hours at home, and by late 2023 that share was ~20% [25†L85-L89] [29†L182-L190] . (For example, StatCan reports ~41% WFH in Apr 2020;  $\approx$ 30% by Jan 2022;  $\approx$ 20% by Nov 2023 [25†L85-L89] .) In May 2021, about 24% of Canadians “usually” worked at home (up from 7.1% in 2016) [27†L109-L112] . By mid-2023 it was ~20%.

By comparison, US levels of regular WFH are somewhat higher: U.S. data show 35–40% of employees working remotely (at least some days) by 2022 [46†L48-L52] . European rates remain lower – Eurostat data in 2021 show only ~12% of EU workers “usually” WFH [44†L27-L34] . (For example, Finland topped EU at 25%, but many countries are <10% [44†L27-L34] .) Thus **Canada/US ~20–40% WFH vs EU ~10–20%** in recent years.

An approximate timeline:

- **2019:**  $\approx 7\%$  in Canada (pre-pandemic baseline).
- **2020:** Sharp rise –  $\sim 40\%$  in April 2020 [25†L85-L89] .
- **2021:** Roughly 25–30%. (StatCan:  $\sim 30\%$  Jan 2022 [25†L85-L89] , implying similar late-2021.)
- **2022:**  $\sim 23\text{--}24\%$  (year average). (For example, Dec 2022  $\sim 22.1\%$  [29†L178-L181] .)
- **2023:**  $\sim 20\%$  (Nov 2023  $\sim 19.7\%$  [29†L189-L194] ).
- **2024:** Estimated  $\sim 15\text{--}20\%$  (e.g. Jan 2024  $\sim 16\%$  in LFS).

(Sources: Statistics Canada Labour Force Survey data [25†L85-L89] [29†L182-L190] .)

### 3. Forecasts to 2030 (Office Occupancy & WFH Scenarios)

**Real-estate/energy-modeling projections:** Most analyses foresee *partial* return-to-office, not full. McKinsey (2023) estimates that **office demand** in the “median city” will be  **$\sim 0\text{--}13\%$  lower by 2030** than in 2019, reflecting longer-term hybrid adoption [36†L618-L627] . (In a survey of 9 global cities, McKinsey projected demand drops of  $\sim 16\text{--}20\%$  in tech hubs, small increases in a few cities [38†L28-L36] .) In other words, even with hybrid work, many companies will hold 80–90% of their pre-COVID space. Moody’s cites a McKinsey study: “13% less demand for office space in the median city by 2030” [36†L618-L627] .

Energy or buildings models that incorporate occupancy similarly assume a range of hybrid levels. For example, building simulations often reduce peak occupant densities by a “remote-work factor” to estimate loads (see next section).

**2030 WFH scenarios:** We define three illustrative scenarios for 2030:

- **“Conservative Return” (Low WFH)** – Office usage recovers almost fully. Only  $\sim 15\text{--}20\%$  of workdays are remote by 2030 (i.e. 80–85% in-office attendance). This could reflect strong RTO mandates or structural shifts back to in-person work. It corresponds to only a small additional decline from today’s  $\sim 20\%$  WFH.
- **“Hybrid Normal” (Mid WFH)** – Hybrid stabilizes at moderate levels. Approximately **30%** of workdays are remote (70% in office). This aligns with trends (Stanford projects  $\sim 25\rightarrow 35\%$  WFH by 2035 [42†L125-L132] ) and McKinsey’s median city outlook ( $\approx 13\%$  drop in space). It implies offices at  $\sim 75\text{--}80\%$  of 2019 attendance by 2030.
- **“Fully Hybrid” (High WFH)** – Remote becomes dominant for many.  $\sim 40\%$  of workdays are remote (60% in-office). This assumes continued technology shifts and employee preference. It lies at the high end of forecasts (e.g. Stanford’s 35% by 2035 suggests  $\sim 30\text{--}35\%$  by 2030).

Each scenario implies a corresponding range of occupancy: e.g. if all employees have 5 desk/week, then 20% WFH means 80% desks used on average. We will test these as a single “WFH rate” lever. McKinsey’s finding that  $\sim 0\%$  demand loss median by 2030 [38†L25-L33] corresponds roughly to our “Hybrid Normal” (30% WFH) scenario, while high-remote forecasts (35–40%) drive occupancy down near the “Fully Hybrid” case.

**Canada vs. US/EU:** Canadian projections are broadly similar to the U.S. case. Canadian employers expect to allow significant WFH (e.g. surveys of large firms often target 3 days/week in office), so our scenarios apply equally. EU countries may skew lower (EU WFH was ~12% in 2021 [44†L27-L34] ), so a Canadian/EU note: *Europe's return-to-office has lagged North America's, so Canadian occupancy may recover slightly faster and to a higher level than EU norms, all else equal.*

## 4. Energy Demand Consequences

Hybrid/work-from-home reduces office loads but **not proportionally**. A building left half-empty still uses much of its base energy (HVAC, lighting, servers, etc.). For example, OccuSpace notes that “HVAC and pumps still run like the building’s full [and] lighting often stays on all day” when occupancy dips [55†L109-L117] . Unless systems are dynamically adjusted, a 50% headcount drop might yield far less than 50% energy savings.

Conversely, buildings with occupancy sensors or smart controls can turn off lights and reduce HVAC in empty zones. In one simulation of a “Zero Emission” office building, adding remote days cut annual **energy use intensity (EUI)** up to ~46% [54†L39-L47] . (That case assumed comprehensive controls and two remote days/week.) In practice, partial reductions in peak load and energy can be achieved, but fixed schedules (e.g. minimum ventilation) and plug loads create a baseline demand. Thus **energy modeling should reflect these nonlinearities**: e.g. reduce peak loads and occupancy-driven gains, but retain a nonzero overnight/daytime base.

Load-shape effects: More work-at-home shifts demand from commercial to residential hours. The **office peak** (mid-morning, midweek) flattens when many workers stay home, reducing peak office electricity demand (lighting/compute) but often *raising* grid demand elsewhere. HVAC peaks (cooling/heating) may also shift (colder nights demand heat at home, less at office). Energy models therefore track occupancy schedules carefully.

In sum, studies find ~20–50% occupancy reductions typically yield 10–30% building energy savings, not one-to-one [54†L39-L47] [55†L109-L117] . For example, the Norwegian study showed 2 remote days/week gave ~46% EUI drop [54†L39-L47] (almost linear), but OccuSpace warns that without controls the same headcount drop wastes energy on “empty air” [55†L109-L117] .

## 5. Representing WFH in Energy Models

We treat “WFH rate” as a single **scenario parameter** that scales office occupancy schedules. Concretely, one can reduce the baseline occupancy profile (e.g. 100% presence on Tue/Wed) by the WFH fraction. For example, a 30% WFH rate means only 70% of seats are filled on average. In models (e.g. EnergyPlus or stock building models), this can be implemented by multiplying occupant densities or by omitting occupants on certain days. Some studies explicitly construct “occupancy scenarios” (e.g. 10 scenarios from full-time on-site to extensive remote) [54†L31-L39] . For our forecasts, we will vary the WFH parameter (0–50%) to test sensitivity.

**Table: Year-by-year office occupancy (relative to 2019) and WFH share (Canada).** (Lines labelled “% of pre-2020” are comparisons to Feb 2020 =100%.)

| Year  | Canada WFH share (% usually at home) | US office occupancy (peak day)     |
|-------|--------------------------------------|------------------------------------|
| 2019  | ~7% 【27†L109-L112】                   | 100% (baseline)                    |
| 2020  | ≈40% (Apr 2020) 【25†L85-L89】         | ≈20% (late 2020)                   |
| 2021  | ≈30% (early 2021) 【25†L85-L89】       | ≈35% (peak days)                   |
| 2022  | ≈24% 【29†L169-L177】                  | ≈50% (peak days)                   |
| 2023  | ≈20% 【29†L182-L190】                  | ≈58% (peak midweek) 【17†L210-L218】 |
| 2024  | ≈16–20% (estimate)                   | ≈62% (peak midweek) 【14†L57-L60】   |
| 2025+ | -                                    | ↑ gradually toward 70–80%?         |

(Sources: StatsCan Labour Force Survey 【25†L85-L89】 【29†L182-L190】 ; Kastle/Axios occupancy data 【14†L57-L60】 【17†L210-L218】 .)

**2030 Scenarios (named):** We propose *Conservative Return*, *Hybrid Equilibrium*, and *Fully Hybrid* scenarios. In **Conservative Return (Low WFH)**, WFH stays near 15–20% by 2030 (offices ~80–85% filled). In **Hybrid Equilibrium (Central)**, WFH reaches ~30% (70% in office), reflecting a mature hybrid norm. In **Fully Hybrid (High)**, WFH rises to ~40% (60% in office). These bracket the plausible range implied by surveys and forecasts (e.g. McKinsey’s ~13% space reduction 【36†L618-L627】 and Stanford’s 35% by 2035 prediction 【42†L125-L132】 ).

**Canadian vs. US/EU differences:** Canada’s remote work trajectory mirrors the U.S.: WFH soared to ~40%, then settled ~20%. By 2023, Canadians (20% WFH) were similar to Americans. Europeans lagged (≈12% WFH in 2021 【44†L27-L34】 ). Thus our scenarios (20–40% WFH) are appropriate for North America; Europe may lie toward the lower end (e.g. 10–30% by 2030).

**References:** [Empirical occupancy data and surveys 【14†L57-L60】 【17†L210-L218】 【25†L85-L89】 【29†L182-L190】 ; industry reports on occupancy/utilization 【4†L126-L134】 【20†L295-L304】 ; energy-model studies 【54†L39-L47】 【55†L109-L117】 ; forecasts and analyses 【36†L618-L627】 【42†L125-L132】 【46†L48-L52】 【44†L27-L34】 .